

LongSight: Compute-Enabled Memory to Accelerate Large-Context LLMs via Sparse Attention

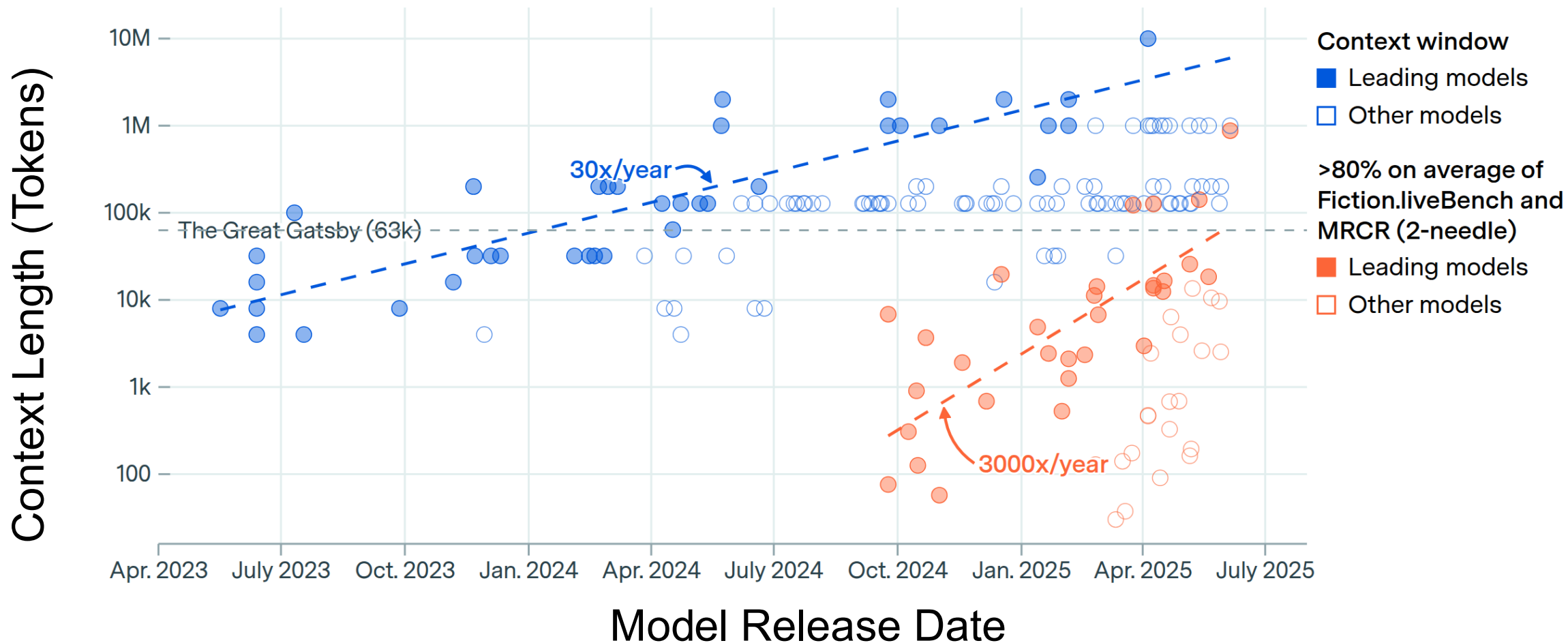
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Cornell University

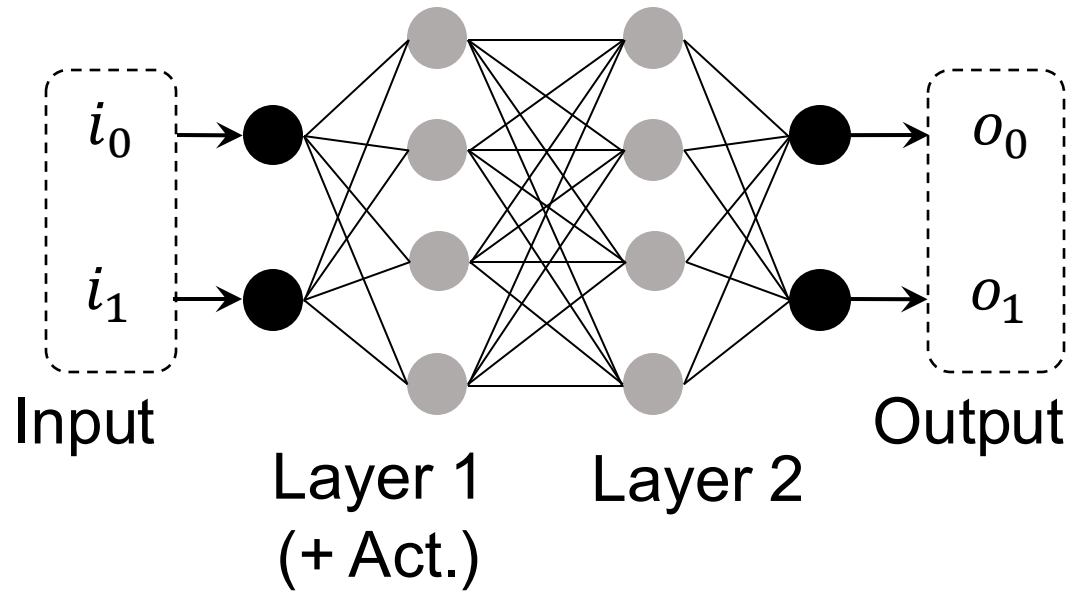
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Electrical and Computer Engineering



Context Lengths Over Time (Epoch AI)



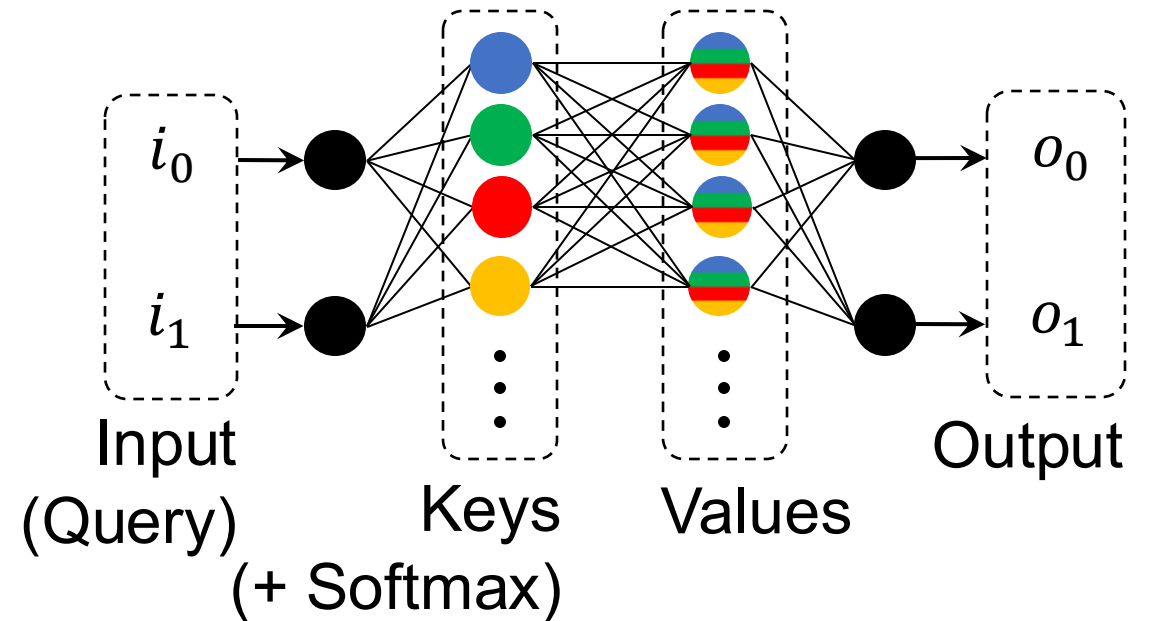
Feed-Forward Network (FFN)



Parameters: $O(D^2)$

Weights: Shared across users

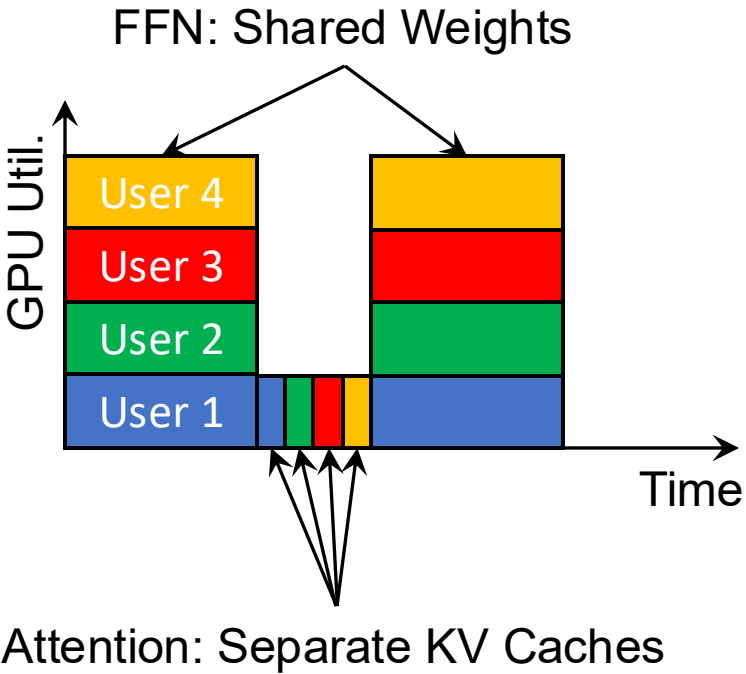
Attention



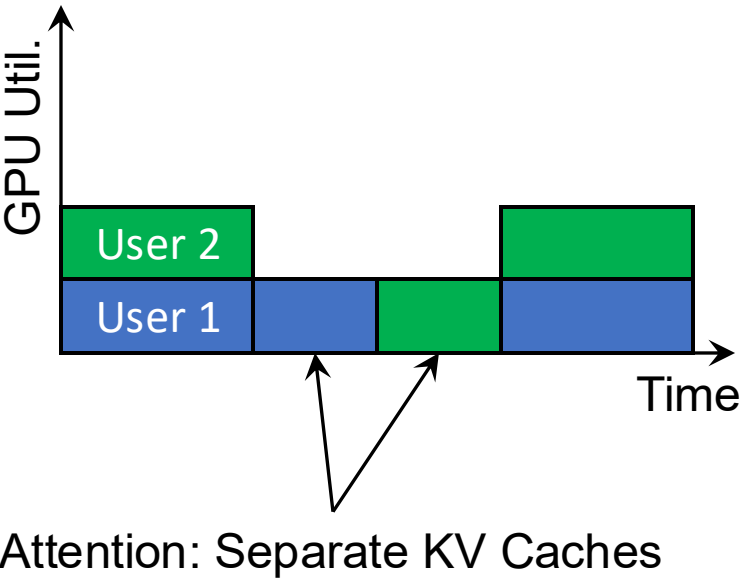
KV Cache “Parameters”: $O(D \times \text{Ctx.})$

Weights: Separate per user

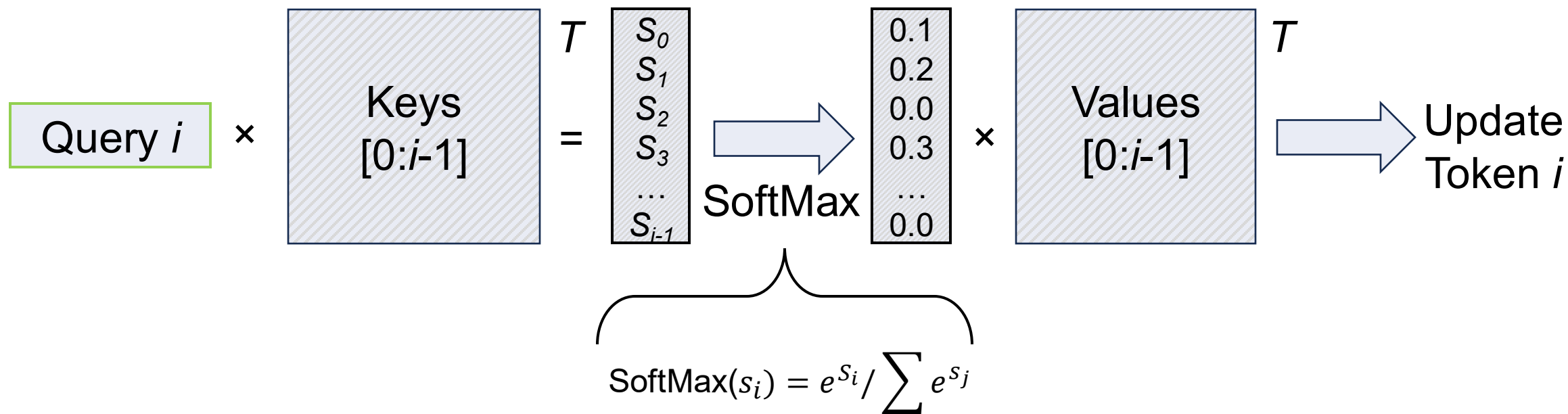
Batched Inference: Short Context

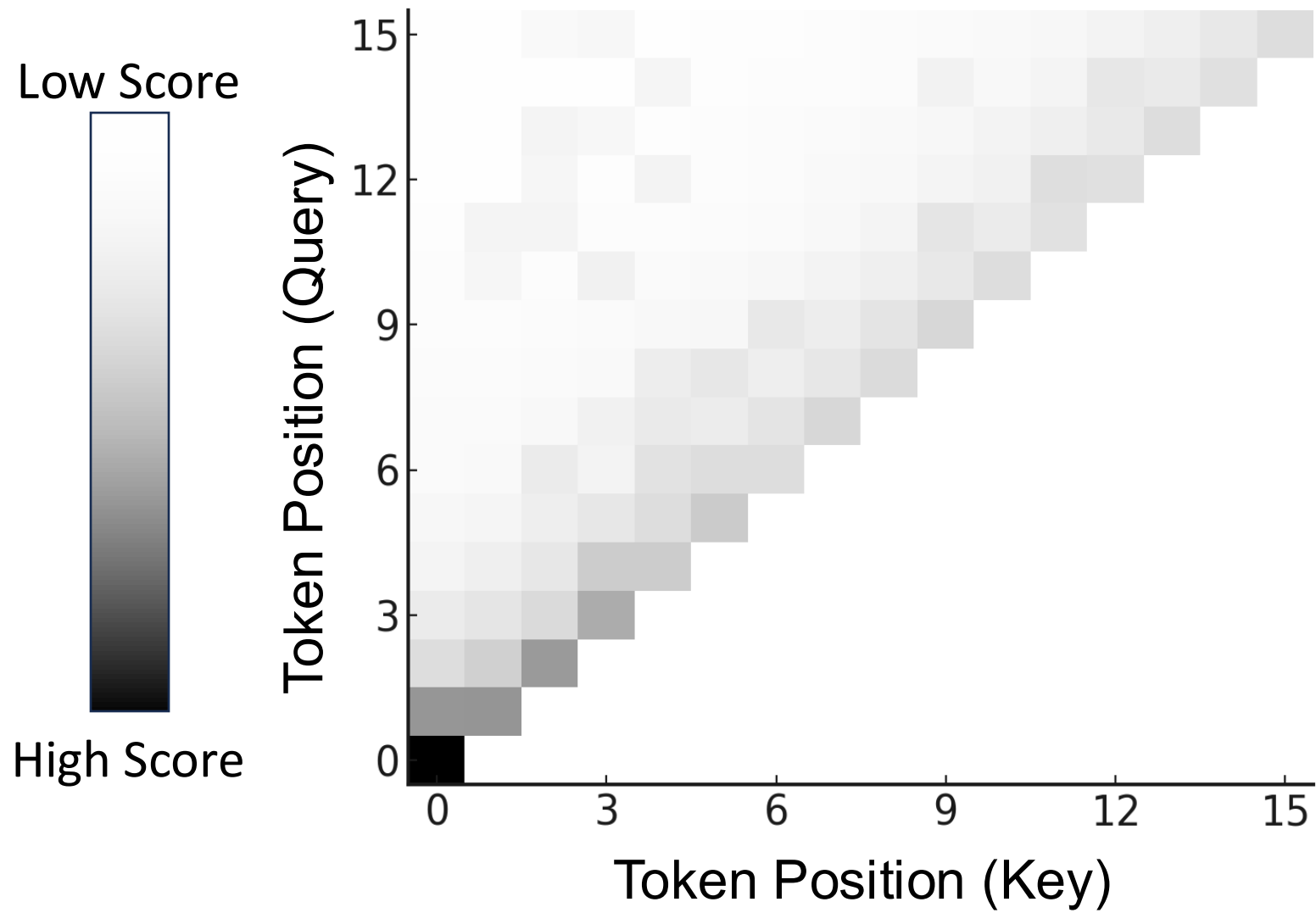


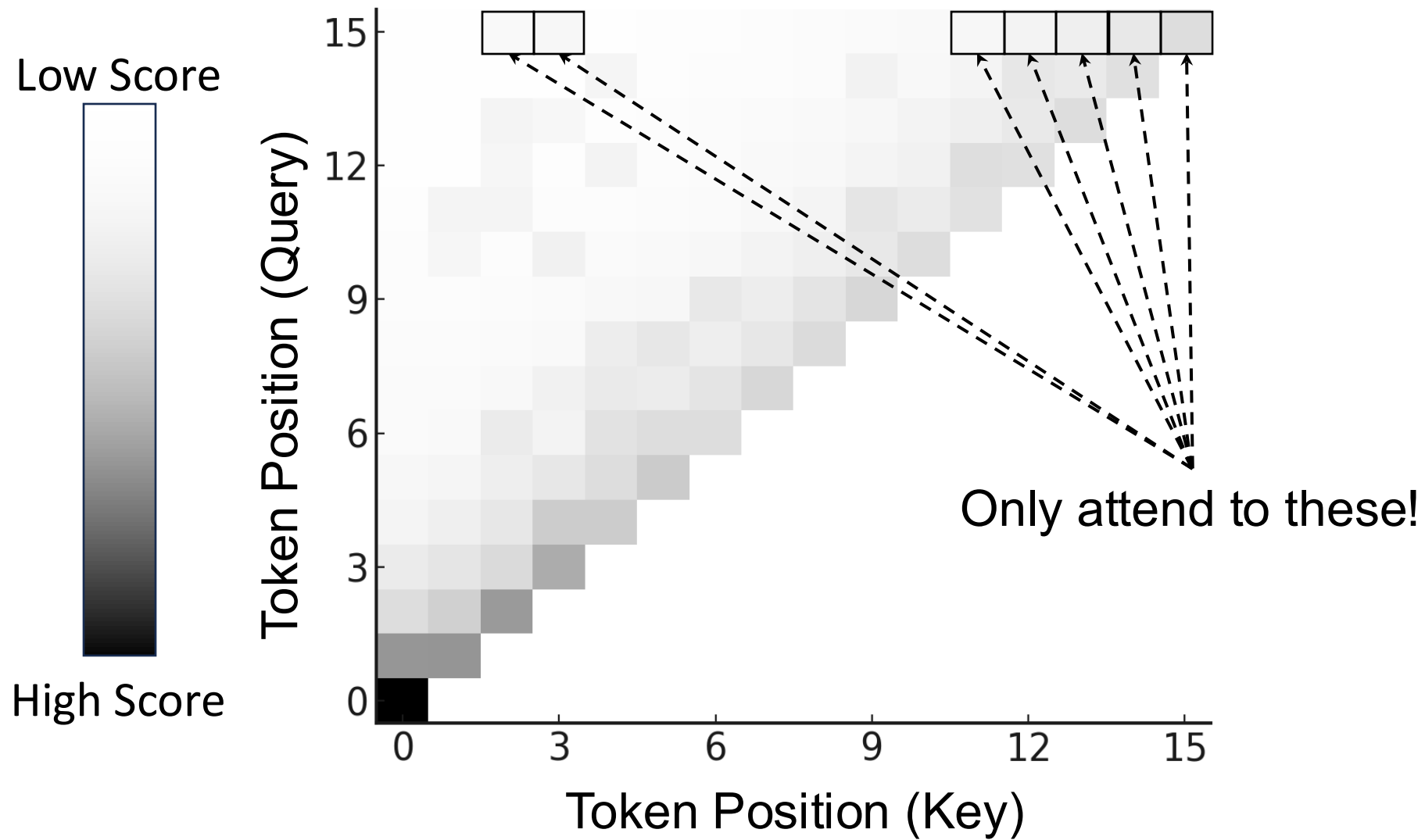
Batched Inference: Long Context



Opportunity: Sparse Attention





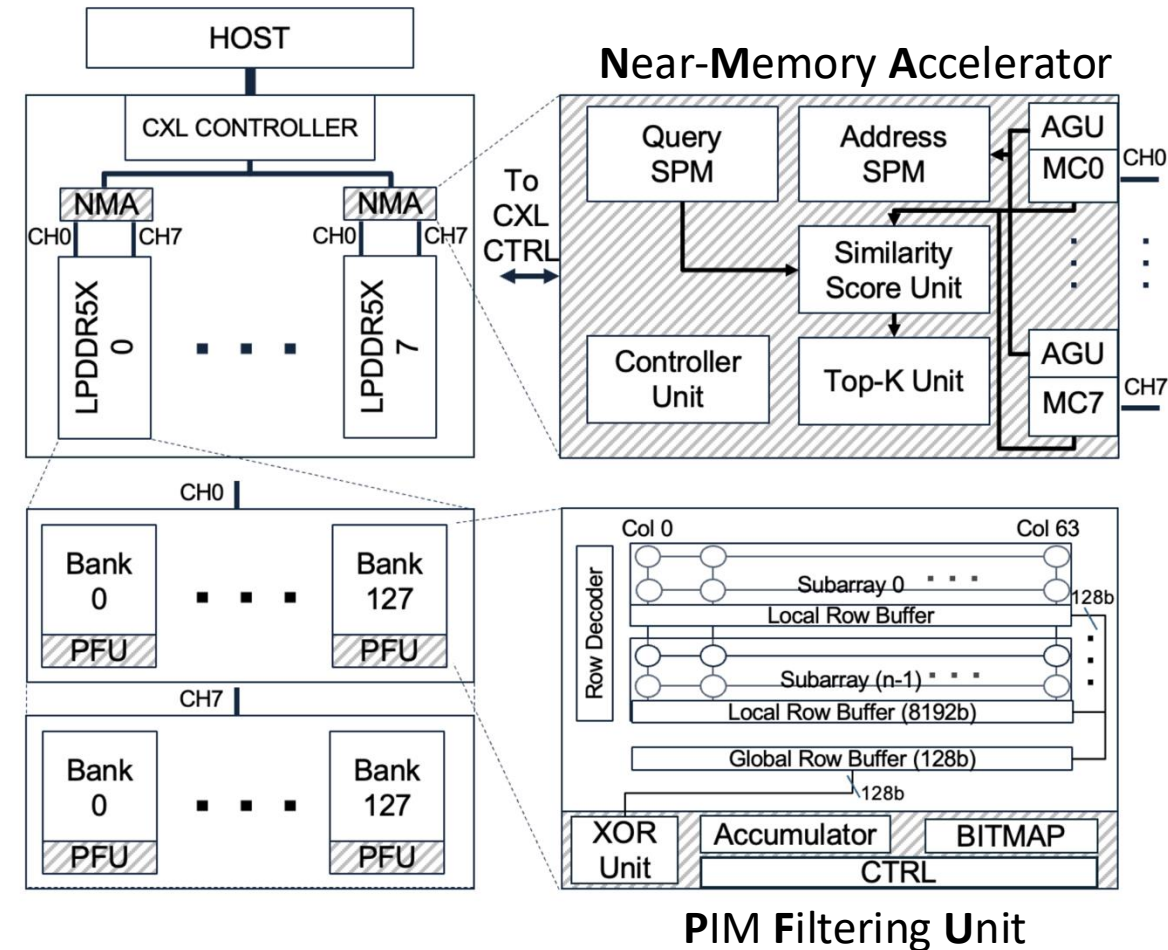


Proposal: Top-K Sparse Attention

1. Store KV cache in a high-capacity, compute-enabled CXL memory
2. Filter out less relevant keys **inside DRAM**
3. Find Top-K Keys with the highest attention score **near DRAM**
4. Ship the **Top-K Keys and Values** to GPU for Attention

DReX (ISCA 2025*)

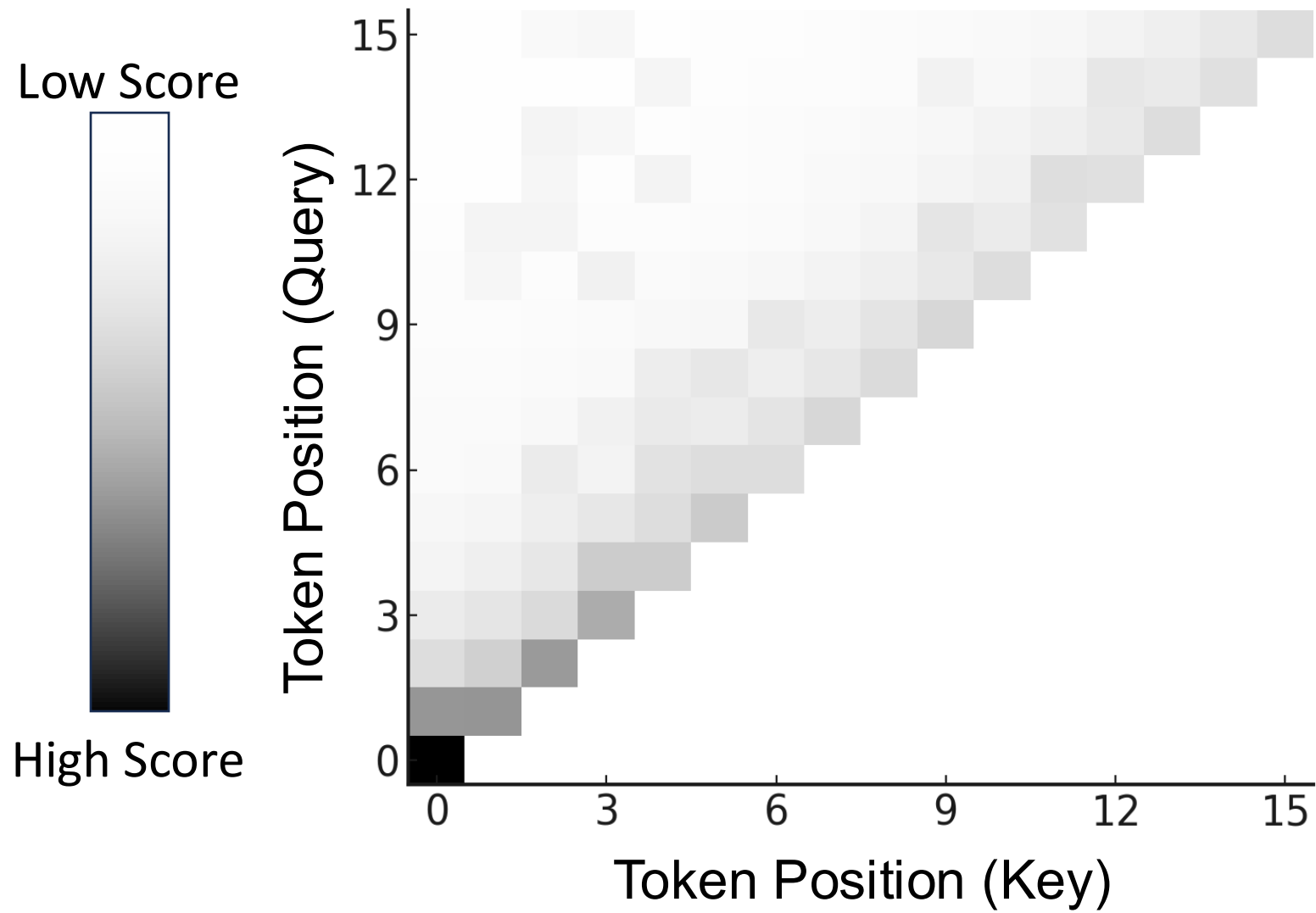
- **Dense Retrieval Accelerator (DReX)**
 - Compute-enabled CXL-memory expander
- 512 GB Capacity
- Sign Concordance Filtering (**SCF**)
 - Filter vectors based on sign bits
- PIM Filtering
 - Run SCF inside DRAM
- Near Memory Ranking -> Top-k retrieval
 - Full-precision similarity score near DRAM

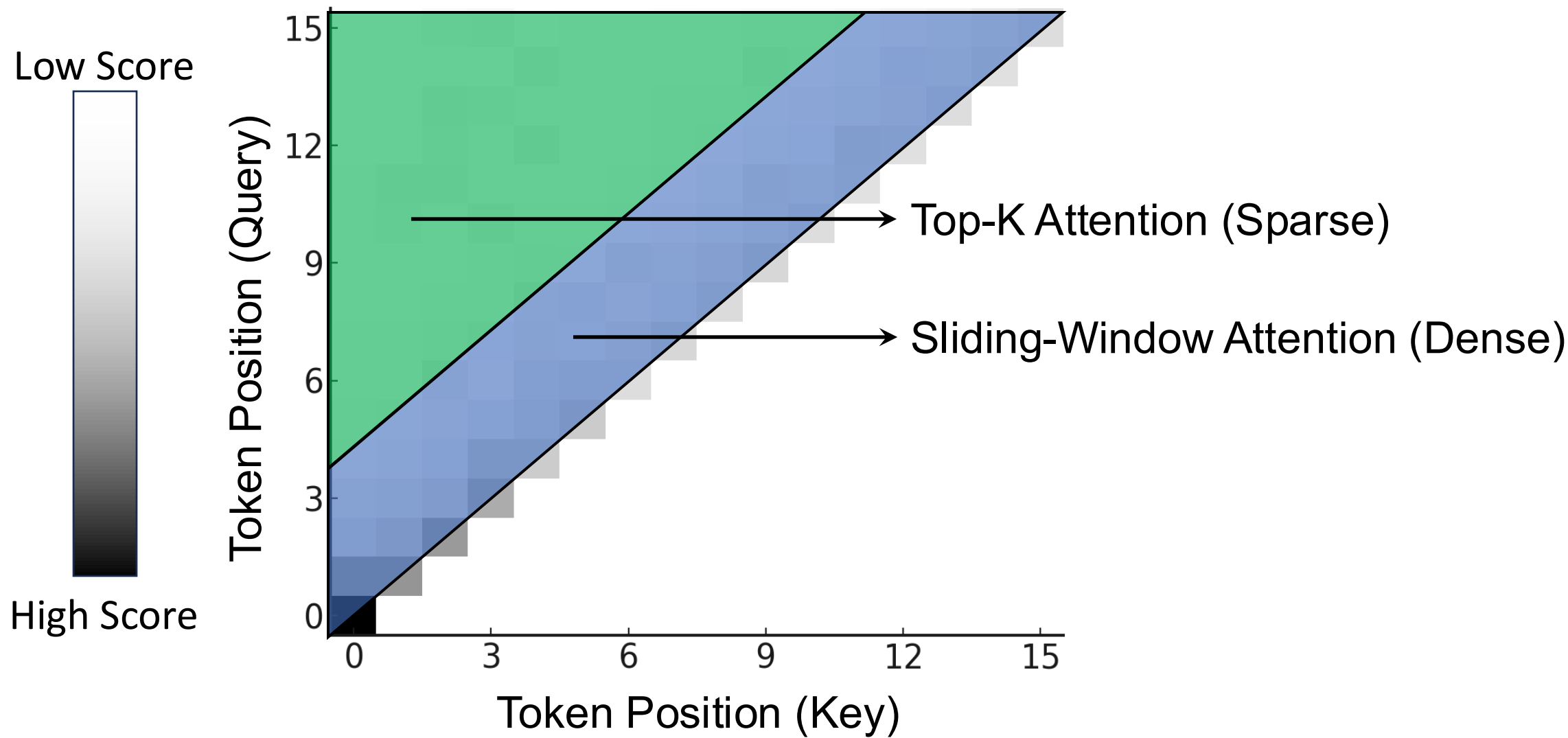


* D. Quinn, E. Yucel, M. Prammer, Z. Fan, K. Skadron, J. Patel, J. Martínez, and M. Alian, "DReX: Accurate and Scalable Dense Retrieval Acceleration," ISCA 2025

Reimagining Attention for DReX

1. Combine SCF with sliding-window attention: **balances load with GPU and improves SCF sparsity**

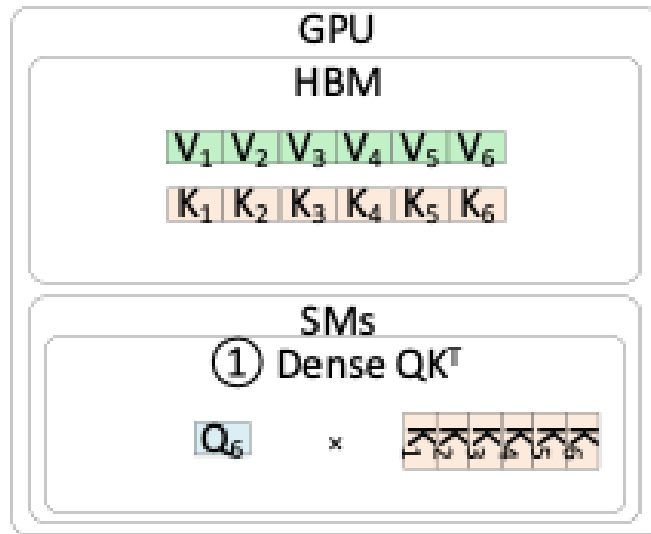




Reimagining Attention for DReX

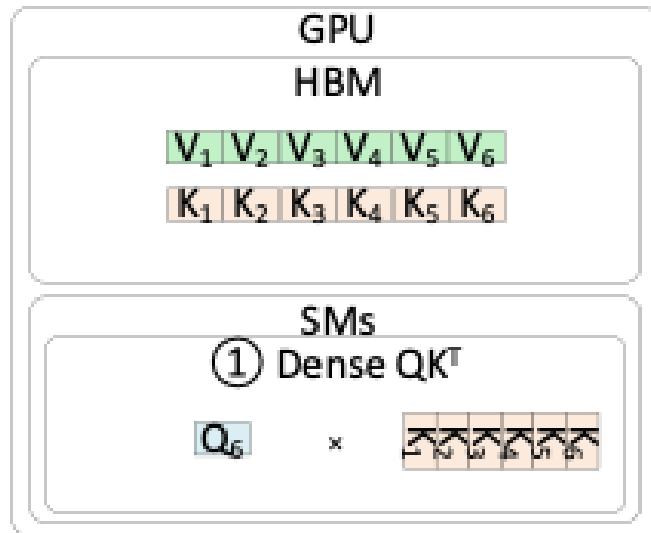
1. Combine SCF with sliding-window attention: **balances load with GPU and improves SCF sparsity**
2. Optimize Queries and Keys via ITQ: **further improves SCF sparsity** by rebalancing vectors

Conventional GPU-Only Attention

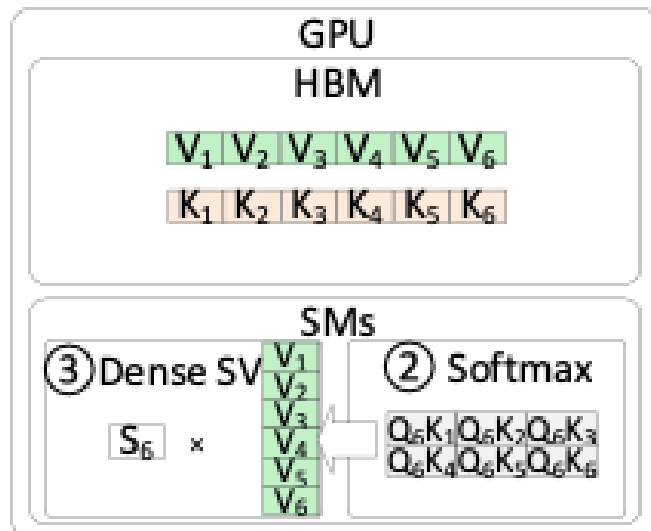


Step 1. Dense QK^T for all tokens

Conventional GPU-Only Attention

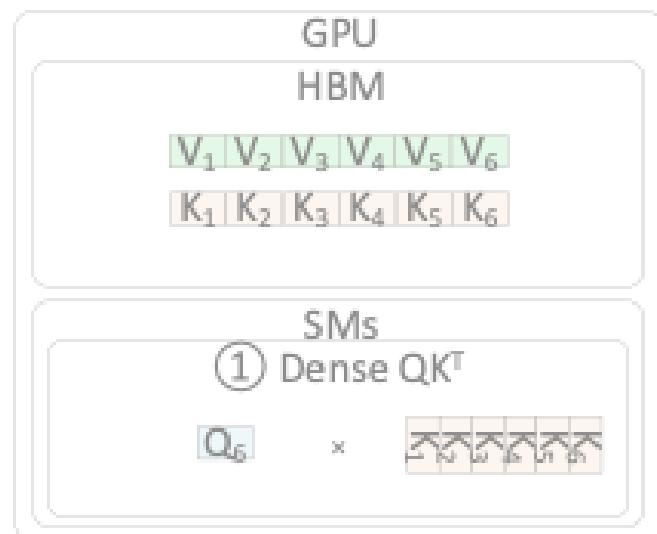


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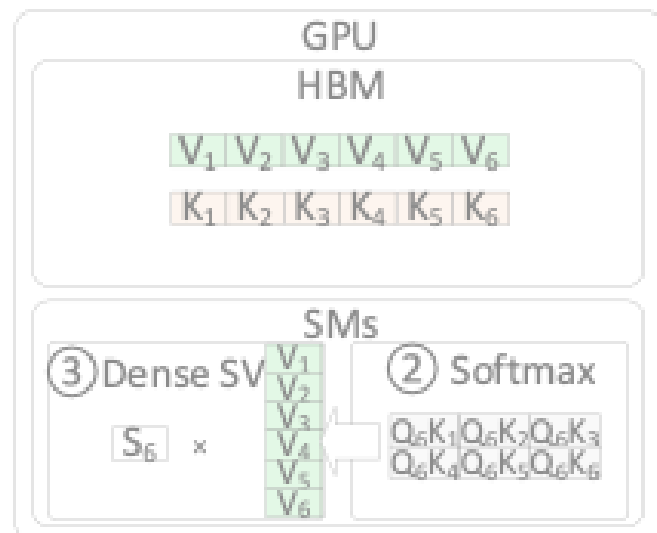


Step 2. Softmax & Dense SV

Conventional GPU-Only Attention

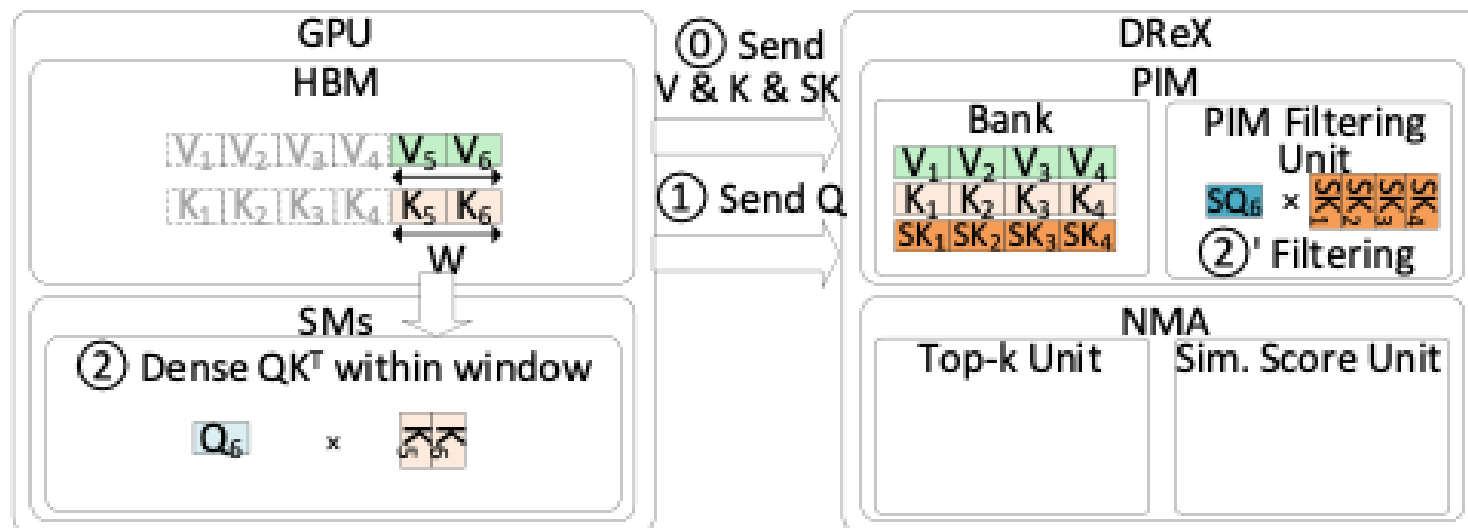


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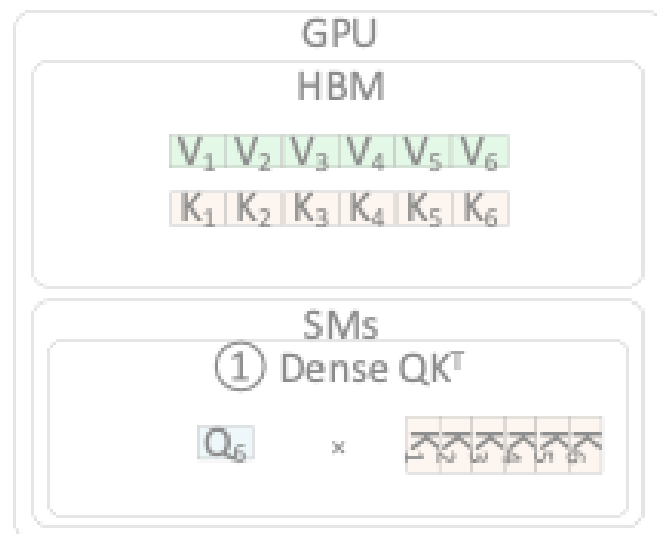
Step 2. Softmax & Dense SV

Our Approach (LongSight Attention)

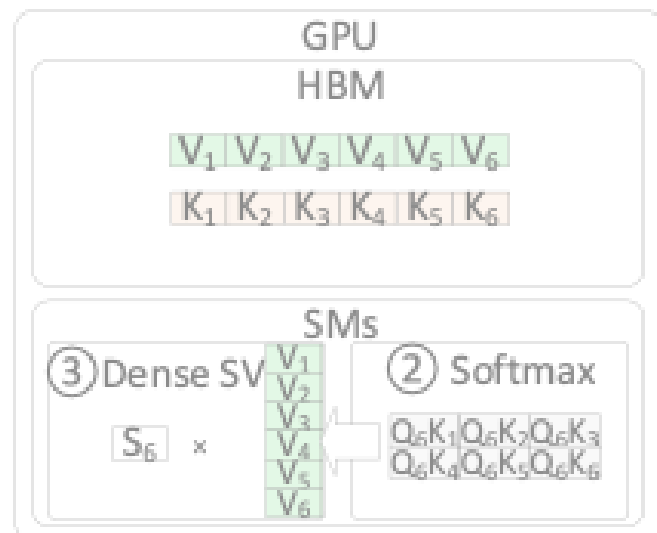


Step 1. GPU : Dense QK^T within Window / DReX : Filtering

Conventional GPU-Only Attention

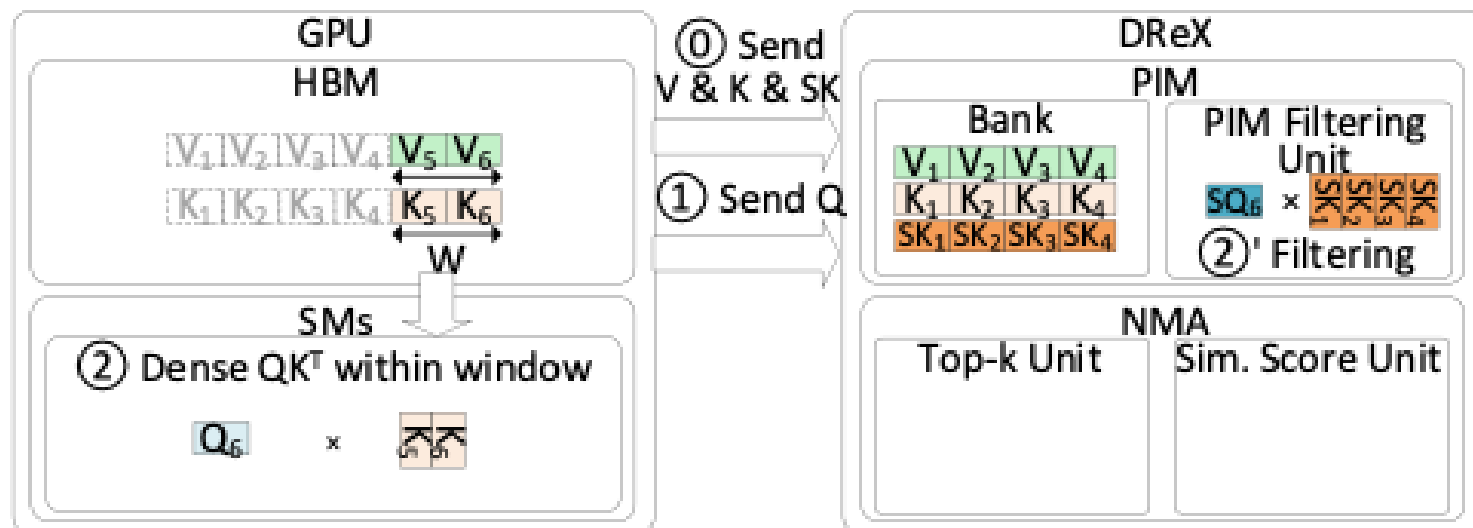


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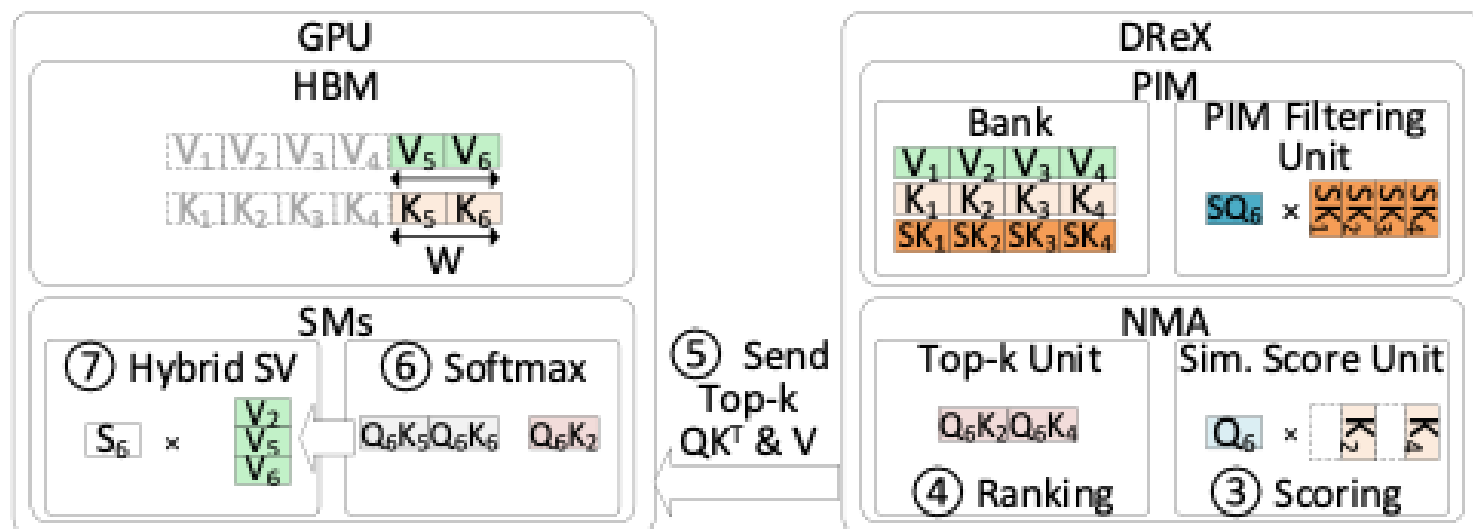


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Our Approach (LongSight Attention)



Step 1. GPU : Dense QK^T within Window / DReX : Filtering



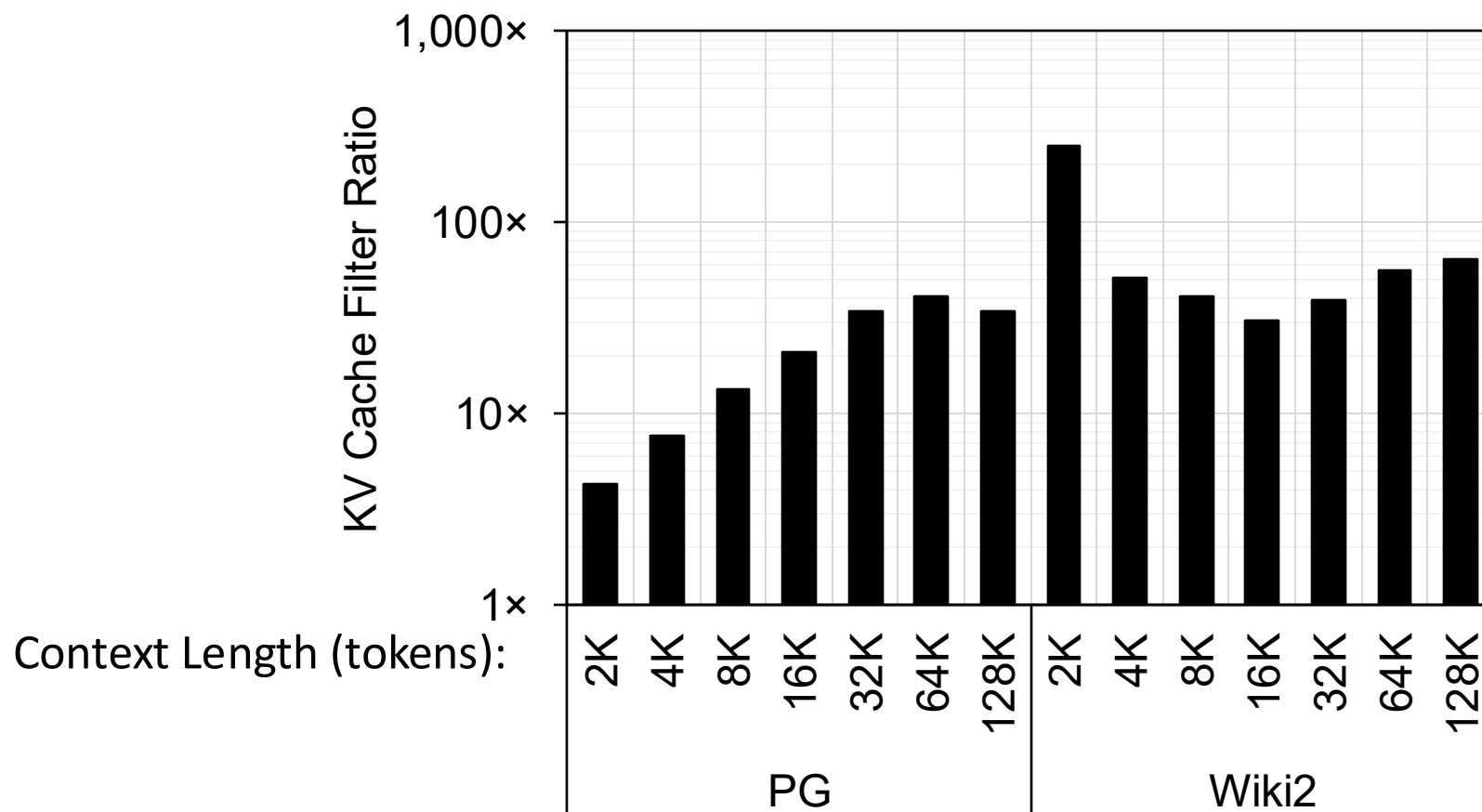
Step 2. GPU : Softmax & Hybrid SV / DReX : Scoring & Ranking

Methodology

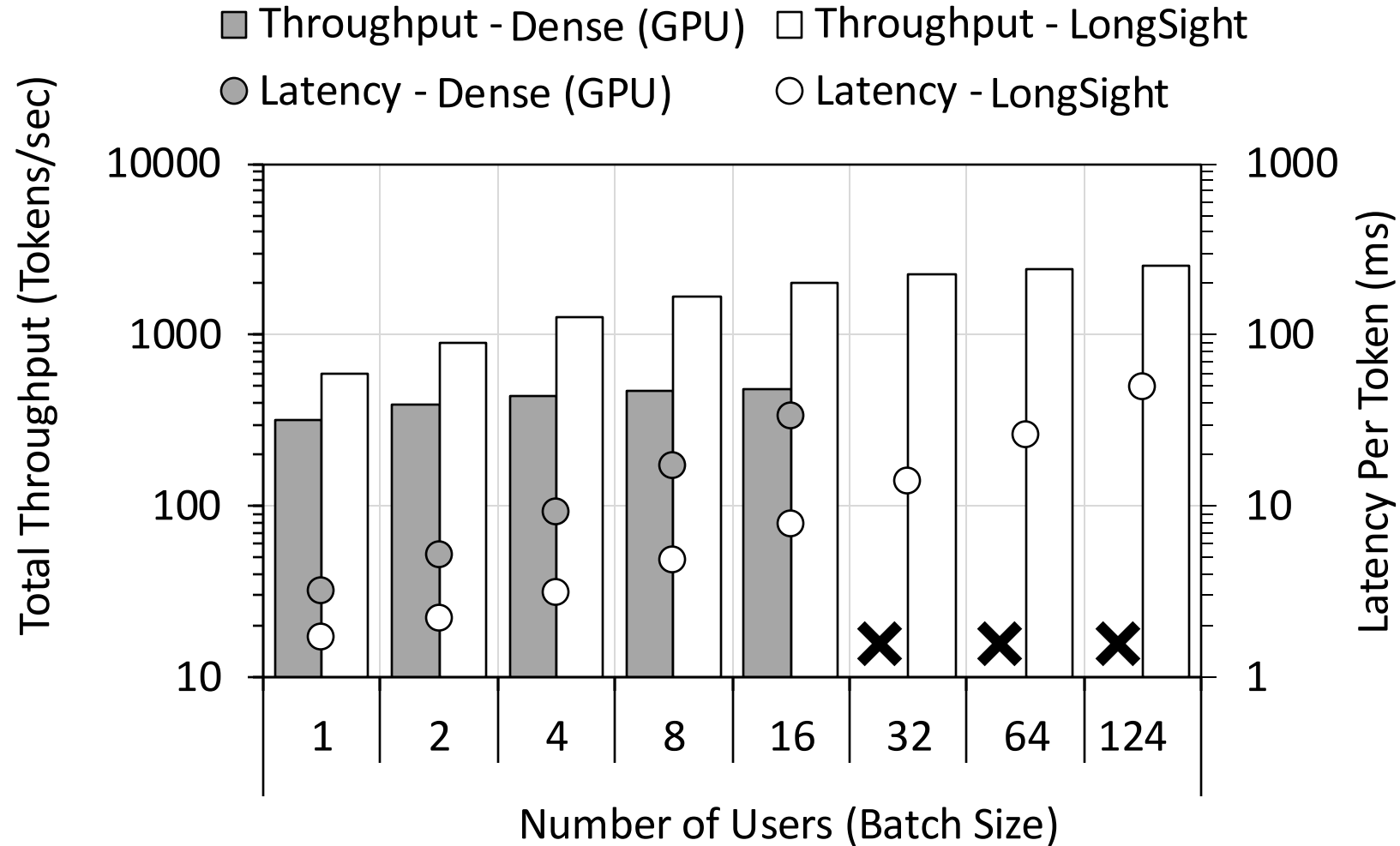
- Models: Llama-3-1B
- Datasets: Project Gutenberg (PG) and Wikipedia (Wiki2)
- Context lengths: Up to 128K (evaluated), 1M (projected)
- We evaluate decode phase (no prefill)
- End-to-end timing simulator:
 - Non-attention components (FFN, etc.): GPU-calibrated
 - Dense Attention Baseline: GPU-calibrated
 - Sparse Attention: RTL-validated analytical model for DReX
 - CXL interface: Remote accesses in two-socket Xeon server

Algorithmic Results: Hybrid Attention

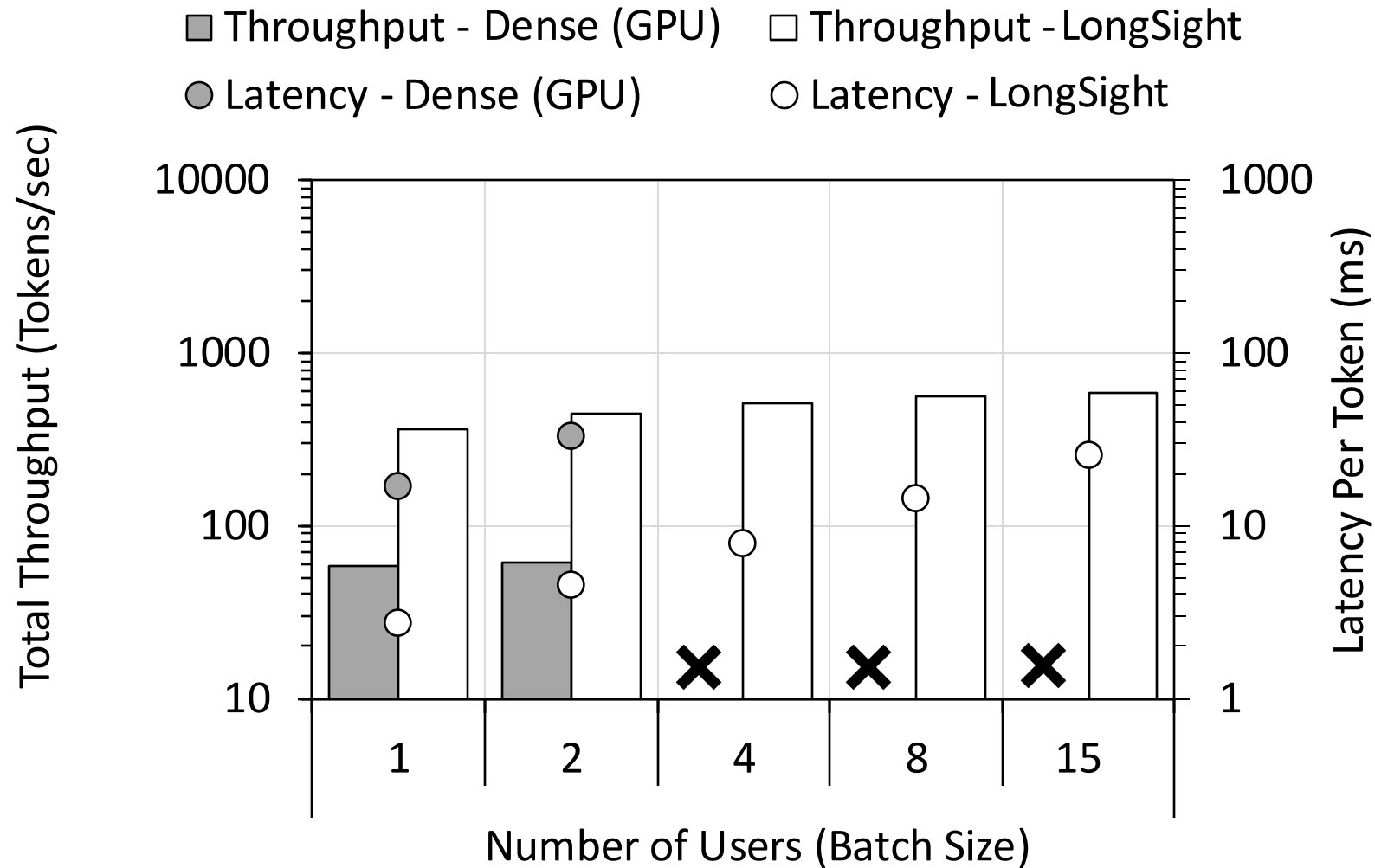
- Perplexity is within 5% of full dense attention.
- $K \leq 1024$



Results: Llama-3-1B – 128K Token Context



Results: Llama-3-1B – 1M Token Context



Contributions

1. LongSight leverages DReX to sparsify long-context attention
2. LongSight's hybrid attention balances load between DReX and GPU and improves robustness
3. LongSight efficiently disaggregates long-context attention, expanding system capacity and curtailing underutilization